

Polynomial Regression Assignment Report

Name: Hasini | Roll number: BT2024113

1. Objective and approach in brief

The task is to predict a continuous target y from the inputs for two personalised datasets using only polynomial regression: var1 has six inputs (degree at most 10) and var2 has three inputs (degree at most 20). Each dataset has 1000 training rows and 1000 test rows. The main risk is choosing a degree that is too low (underfitting) or too high (overfitting), so the whole method is built around choosing the degree and the regularisation strength in a principled way using cross-validation, instead of guessing them.

Final models. var1: degree 5, Ridge $\alpha = 10$ (461 polynomial terms). var2: degree 10, Ridge $\alpha = 1$ (285 polynomial terms). Cross-validated R^2 is 0.949 for var1 and 0.994 for var2.

2. Data overview

	var1	var2
Training / test rows	1000 / 1000	1000 / 1000
Number of inputs	6 ($x_1 \dots x_6$)	3 (x_1, x_2, x_3)
Maximum degree allowed	10	20

Two observations shape the method. First, all inputs already lie in $[-1, 1]$, so the polynomial terms stay well scaled. Second, a large share of input values (about 31% in var1 and 25% in var2) sit exactly at -1 or 1, which suggests the inputs were clipped to this range when the data was generated. This leaves fewer points in the interior and makes the corners of the input space important.

There are no missing values.

3. Method

3.1 Model. For each candidate degree d , the inputs are expanded into all polynomial terms whose powers sum to at most d (scikit-learn PolynomialFeatures, no bias column). The terms are standardised (zero mean, unit variance) using training-fold statistics only, and a linear model is fitted with Ridge regression (L2 penalty). The model is still a polynomial regression; the penalty only shrinks the coefficients.

3.2 Why Ridge is needed. With 1000 samples, var1 at degree 10 has 8007 terms and var2 at degree 20 has 1770 terms. Ordinary least squares is then unstable or not even uniquely defined, and the high-order terms are strongly correlated with each other. The Ridge penalty (strength α) keeps the coefficients small and the curve smooth. A small α behaves like plain least squares (overfitting risk); a large α flattens the curve (underfitting risk).

3.3 Selecting degree and alpha together. Degree and α interact, because a higher degree usually needs a larger α . So I searched them jointly with GridSearchCV: every degree from 1 up to the allowed maximum, combined with α in $\{0.001, 0.01, 0.1, 1, 10, 100, 1000\}$. Each pair was scored by 5-fold cross-validation (shuffled, random seed 42) using mean squared error on the held-out fold. The pair with the lowest mean CV MSE was selected and then refitted on all 1000 training rows to predict the test file.

3.4 Why cross-validation. The test targets are hidden, and training error always keeps falling as the degree grows, so it cannot be used to choose the degree. Held-out error does start rising when the model overfits, which is exactly the signal needed.

4. Results for var1 (six inputs)

CV MSE falls sharply from degree 1 to 5 (9.79 to 0.55), which is clear underfitting at low degrees. It is lowest at **degree 5** and then rises steadily as the degree increases (1.09 at degree 10), which is overfitting: the model has thousands of terms but only 1000 rows and a noisy target.

Degree	best alpha	CV MSE	Std of fold MSE	CV R2
1	100	9.7916	0.4349	0.0885
2	10	3.0964	0.2528	0.7119
3	10	1.0063	0.0485	0.9063
4	10	0.7221	0.0495	0.9329
5	10	0.54934219	0.0327	0.9487
6	100	0.6510	0.0579	0.9394
7	100	0.6855	0.0309	0.9361
8	100	0.7964	0.0678	0.9259
9	100	0.9458	0.1153	0.9119
10	100	1.0921	0.1680	0.8983

5. Results for var2 (three inputs)

Here the curve behaves differently. CV MSE drops steeply up to degree 8 (35.5 at degree 1, 0.262 at degree 8) and then becomes almost flat from degree 8 to 12 (0.251 to 0.268 for degrees 8 to 14). The minimum is at **degree 10** with alpha = 1, CV MSE 0.251. After degree 12 the error creeps up (0.300 at degree 20), so the maximum allowed degree of 20 is not needed.

1	10	35.4681	5.8253	0.2221
2	1	20.8346	2.0853	0.5375
3	0.001	11.2087	2.0533	0.7476
4	0.001	3.4969	0.4695	0.9231
5	0.1	1.6231	0.3733	0.9643
6	0.01	0.6046	0.0286	0.9864
7	0.1	0.3570	0.0840	0.9921
8	0.1	0.2622	0.0302	0.9942
9	0.1	0.2675	0.0351	0.9941
10	1	0.25113748	0.0220	0.9944
11	1	0.2559	0.0239	0.9943
12	1	0.2555	0.0198	0.9943
14	1	0.2682	0.0228	0.9940

20	10	0.3003	0.0332	0.9933
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6. Rationale for the chosen degrees

- **var1, degree 5.** The minimum is clear and the fold-to-fold spread is small (0.033), so it is clear that lower degrees underfit and higher degrees overfit. Alpha = 10 was best; for degrees 6 and above the best alpha moves to 100, showing that the model needs more shrinkage as it gets more flexible.
- **var2, degree 10.** Degrees 8 to 12 differ by less than about 0.012 in CV MSE, which is comparable to the fold-to-fold noise (standard error about 0.01), so they are statistically almost the same. The residual standard deviation of about 0.50 at the chosen model matches this plateau and looks like the noise level of the data, so no polynomial can do much better. I kept the degree with the lowest CV MSE (10). Degree 8 would be the simplest choice with practically the same accuracy.
- **Not using the maximum degree.** The assignment says the true polynomial can be of degree up to 10 and up to 20, but the best predictive degree is lower for var1. With limited, noisy data, a smaller model predicts better than one that matches the highest possible true degree.

7. Final models and performance

Selected degree	5	10
Selected Ridge alpha	10	1
Polynomial terms	461	285
CV MSE (5-fold)	0.5495	0.2509
CV R2 (5-fold)	0.9491	0.9945
Std of CV residuals	0.741	0.501
Training MSE (optimistic)	0.1626	0.1695
Training R2 (optimistic)	0.9850	0.9963

8. Other techniques, checks and limitations

- **Regularisation instead of plain least squares**, with alpha tuned jointly with the degree (Section 3).
- **Standardising the polynomial terms** so that the Ridge penalty treats all terms fairly. Scaling uses training-fold statistics only, so no information leaks from the validation fold.
- **Extrapolation caution.** The var1 test predictions range from -10.5 to 17.1, slightly beyond the training targets (-9.9 to 13.0). This happens at corner points of the input space where high-order terms grow quickly and few training points exist. It is a known weakness of high-degree polynomials and one more reason to prefer the lower degree 5 for var1.
- **Limitations.** The degree and alpha were chosen on the same cross-validation scores, so the reported CV MSE is slightly optimistic. The alpha grid uses steps of one power of ten, so a finer grid could give a small further improvement (for var1, a finer search suggested alpha near 30 with CV MSE about 0.53 instead of 0.55). Only Ridge-regularised polynomial regression was used, as required.

9. Reproducibility

All model selection, final training and prediction generation are implemented in a single Python script. The script generates the two required prediction files.”

GitHub repository: [Hasini-Samudrala/Polynomial-Regression](https://github.com/Hasini-Samudrala/Polynomial-Regression)